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Geological interpretation of palaeomagnetic results from Devonian rocks of Hoy, Orkney

K. M. STORETVEDT[1,2] AND A. H. MELAND[1]

[1] *Institute of Geophysics, University of Bergen, Allegt. 70, 5000 Bergen, Norway*
[2] *Dept. of Earth Sciences, Memorial University of Newfoundland, Canada*

SYNOPSIS

The palaeomagnetic record of Devonian Volcanics of Hoy (~370 Ma), occurring at the base of the Hoy Sandstone (Upper Devonian), exhibits coherent groups of remanence directions. An original shallow magnetization axis, repeatedly reported from studies of Old Red Sandstone rocks of Caithness, is confirmed, but the data suggest also that the tectonic unconformity between the Middle Devonian strata (upon which the volcanics rest) and the Hoy Sandstone originated at around the time of the volcanic activity. A secondary magnetization of inferred Permian age has been uncovered from the underlying Middle Devonian sediments (Lower Eday Sandstone). This overprinted component of magnetization, which is a pronounced feature in various Caithness rocks, is reported here for the first time for Orkney rocks. The Permian overprinting event is held responsible for the demonstrable loss of argon in the Devonian volcanics, rendering also the K/Ar ages for other igneous rocks in the region highly suspect.

INTRODUCTION

In recent years quite a few palaeomagnetic studies have been undertaken on rocks from the Orcadian Basin (for example, Storetvedt and Petersen 1972; Tarling *et al.* 1976; Turner 1977; Turner *et al.* 1978; Storetvedt *et al.* 1978; Storetvedt and Carmichael 1979; Storetvedt and Torsvik 1983). Complex multiphase magnetizations have been uncovered in most of these cases, and for various Caithness rocks a substantial fraction of the original Devonian remanence appears to have been erased during Permo-Mesozoic remagnetization. On the whole, the total palaeomagnetic record has gradually become better understood and as a consequence a number of potential applications of the results to local geological problems can now be envisaged.

In the present contribution we give an account of the magnetization history of Devonian volcanics and baked Devonian sediments from Hoy, Orkney. In a previous study of the Orkney volcanics (Storetvedt and Petersen 1972) it was found that these rocks (including partly the exposures dealt with in the present study) provide a striking example of extensive low temperature oxidation of the Fe-Ti oxides. Some lava specimens exhibit traces of high temperature deuteric oxidation, but these features are always strongly overprinted by subsequent low

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temperature processes. Alteration products of the latter constitute either: (a) a fine-grained mineral assemblage of granulation texture (Ade-Hall *et al.* 1968) which do not attract magnetic colloid, probably consisting mainly of rutile; or (b) a fine-grained intergrowth of different minerals such as rutile, haematite and titanium-poor spinel (probably titanomaghemite). The presence of the latter two minerals was also clearly demonstrated by thermomagnetic measurements. The original titano-magnetite (TM) grains of the Devonian tuff layer of Hoy have undergone a complete oxidation in that all TM particles now tend to consist of haematite and rutile. The more severe oxidation state of the tuff, as compared to that of the lavas, is in part attributed to its fairly porous character. Thus, given rocks like the Orkney volcanics, one must anticipate a lengthy magnetization record of unknown duration. The palaeomagnetic structure of such rocks would normally be a complex one (as was also demonstrated in the earlier study), but provided individual palaeomagnetic components can be successfully extracted, important information on the geological history of the region might be deduced. In view of the progress in palaeomagnetic instrumentation and techniques over the past 15 years, it was decided to re-assess the palaeomagnetism of the Hoy volcanics with the hope of achieving more information than that obtained in the original study. The present investigation is believed to have been quite successful in this respect, and the results appear to have interesting implications for tectonic and age problems.

General geology and sampling

At the base of the Hoy Sandstone of Upper Old Red Sandstone age (Sedgwick and Murchison 1828; Wilson *et al.* 1935; Halliday *et al.* 1977) a volcanic horizon is exposed at a few localities notably in the northern part of the island (Fig. 1). The volcanicity was of explosive character and an ashy deposit with thicknesses ranging upward to *c.* 15 m was laid down prior to lava outpouring. In the various lava outcrops only one flow is recognized at any one locality, and the petrography is always that of an olivine basalt with phenocrysts of olivine, pyroxene and feldspar (Wilson *et al.* 1935). Faults occur between the separate igneous exposures, and from field evidence it is impossible to be certain of the precise stratigraphical relation between the outcrops.

In northern Hoy the volcanics rest on Middle Old Red Sandstone strata and between these sediments and the overlying Hoy Sandstone there is a fairly marked tectonic unconformity. Apart from the Old Man of Hoy exposure, which shows evidence of some erosion of the lava prior to deposition of the Hoy Sandstone, the slaggy and reddish upper part of the flow(s) is preserved. This suggests that there is not a substantial time interval between volcanism and deposition of the Hoy Sandstone. ^{40}Ar–^{39}Ar stepwise degassing analysis of two lava samples indicates an age of around 370 Ma (Halliday *et al.* 1977).

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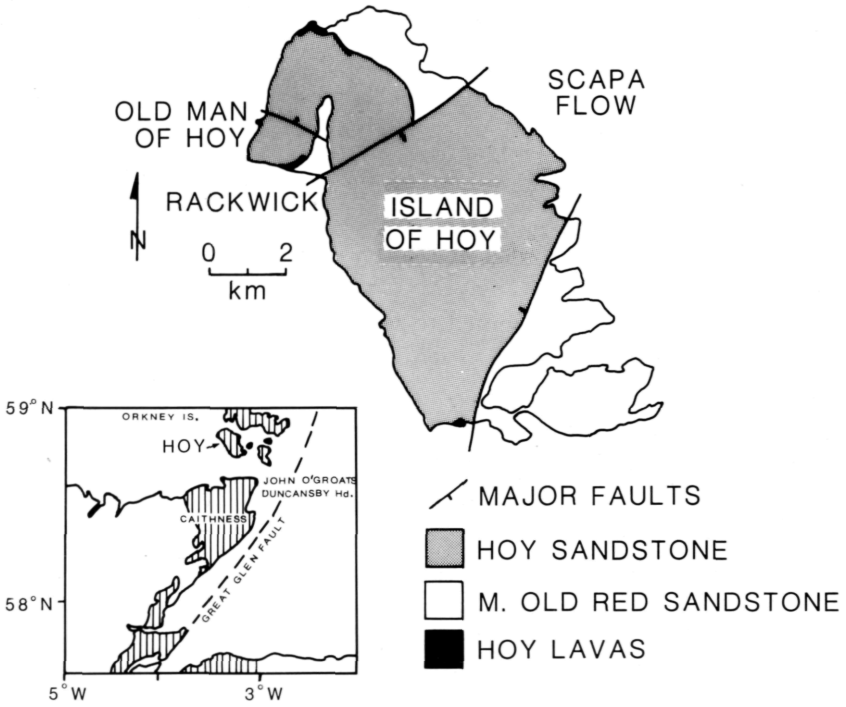


FIG. 1. Geological sketch map of Hoy, Orkney, including main outcrops of the Hoy lavas. Simplified after Halliday *et al.* (1977). The inserted diagram outlines part of the Orcadian basin (shaded) and shows the location of Hoy versus the Great Glen Fault and the areas of Caithness which are referred to in the text.

The present palaeomagnetic investigation deals with the volcanics at Rackwick and at the base of the Old Man of Hoy (Fig. 1). From the latter location the volcanic ash layer (generally found below the lava at each exposure) is missing, and the flow, which in part has pillow structure, rests directly upon the Lower Eday Sandstone forming a pronounced baked contact zone. The sampling sites are distributed throughout the whole thickness of the lava as well as being located at various distances, up to 1 m, away from the lava/sediment contact. At Rackwick sampling was less systematic due to break-down of the drilling equipment, and it was also hampered by the difficulty of obtaining good sites for hand sampling. Further collection details are given in Table 1.

LABORATORY DATA

The remanence measurements have been carried out using partly a Digico Spinner magnetometer and partly a SQUID magnetometer (Cryogenics Ltd.). Thermal demagnetization was performed in a Schonsted TSD-1 demagnetizer, and for alternating field (AF) treatment a two-axes tumbler was used.

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TABLE 1

Rock collection details

Locality	Sites	Sample nos	Other information
Old Man of Hoy	1–4	1–29	Lava forming the base of the Hoy Sandstone (not sampled). Local tectonic dip is *c.* 20 degrees in a WNW direction. Site 4 collected in the chilled margin of lava, *c.* 10 cm above contact, while sites 1–3 are from the interior of flow.
Old Man of Hoy	5 & 6	30–44	Lower Eday Sandstone sampled in the baked contact zone (samples drilled perpendicular to the contact).
Old Man of Hoy	7–10	45–59	Lower Eday Sandstone. With increasing site nos. these have been collected at 20, 30, 50 and 100 cm respectively away from the baked contact.
Rackwick	11–16	60–75	Lava. Site 11 collected by field drill and sites 12–16 by hand sampling.
Rackwick	37–41	175–189	Tuffaceous sediment below Rackwick flow. Sites are situated 7–9 m below lava/tuff contact. Collected by hand sampling. The Rackwick volcanics appear practically undisturbed tectonically.

A total of 76 specimens have been analysed and 41 of these have yielded remanence directions forming distinct groupings. This outcome is surprisingly good taking into account the polyphase magnetizations generally encountered in these rocks, and the fact that in the previous study only 7 specimen directions, from a total of 60 specimens analysed, were accepted for palaeomagnetic consideration. The greater success of the present study in establishing directional endpoints is attributed to the following causes: (a) for samples of the Rackwick lava AF demagnetization was found to be a much better technique than thermal demagnetization which was the only method used in the earlier study; (b) the sediments below the Old Man of Hoy lava, not previously sampled, turned out to be a fairly easy material to work with (lacking the extensive component overlap that is so frequently encountered in the volcanics themselves); and (c) improvement in instrumentation has reduced the experimental/measuring scatter enabling more high field/high temperature end points to be established.

For analytical purposes a combination of stereoplots, orthogonal vector diagrams and vector subtraction have been applied. Specimens that have not been accepted for palaeomagnetic evaluation may be grouped into the following two

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categories: (a) samples that show vectorial movement (often in a complex manner) throughout the full range of demagnetization, and which had too much overlap of their components spectra to enable determination of any individual magnetization vector (in some cases increasing viscosity problems at advanced stages of demagnetization did also prevent the most stable remanence direction from being established); (b) individual specimens from a couple of sites show satisfactory directional stability, but smeared within-site distributions are in favour of unresolved dual-polarity magnetizations.

Old Man of Hoy

The two-polarity remanence of the lava is demonstrated in Figure 2. Samples OR1-A and OR18-A both show high temperature directions regarded as relevant palaeomagnetic field markers. Though specimen OR3-B also provides evidence of the same magnetization structure (coexisting antiparallel components), the results are not considered satisfactory for numerically assessing the palaeomagnetic axis.

The six sites of the underlying sediment (Lower Eday Sandstone) have given very clear-cut results (Fig. 3). A characteristic high temperature direction is in general readily determined above say 400° C. This direction is in complete agreement with that of the normally directed remanence of the lava. The stable

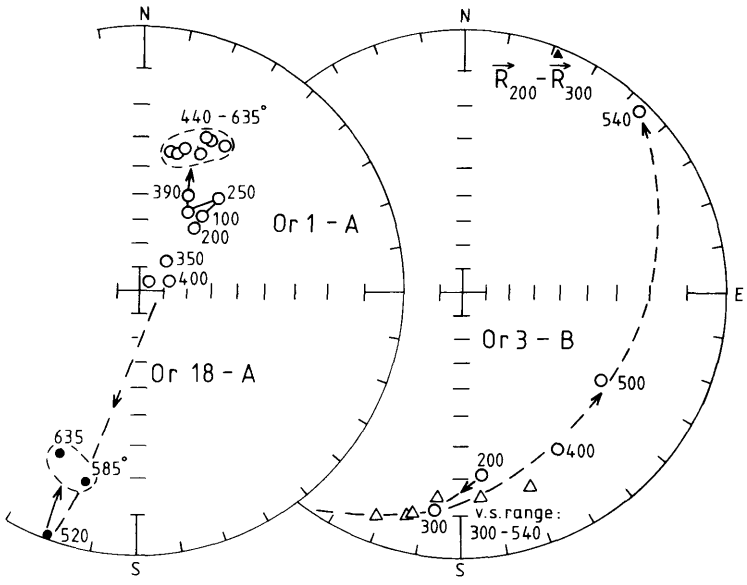


FIG. 2. Examples of coexisting normal and reverse magnetizations in the Old Man of Hoy lava. Closed (open) symbols are downward (upward) pointing directions. Triangles represent vector subtracted directions for specified temperature ranges. Specimen OR3-B displays a spread-out of vector subtracted directions for the northward trending great circle path (300–540° C) signifying overlap of blocking temperature spectra for the suggested antiparallel components. Directions are without tilt correction.

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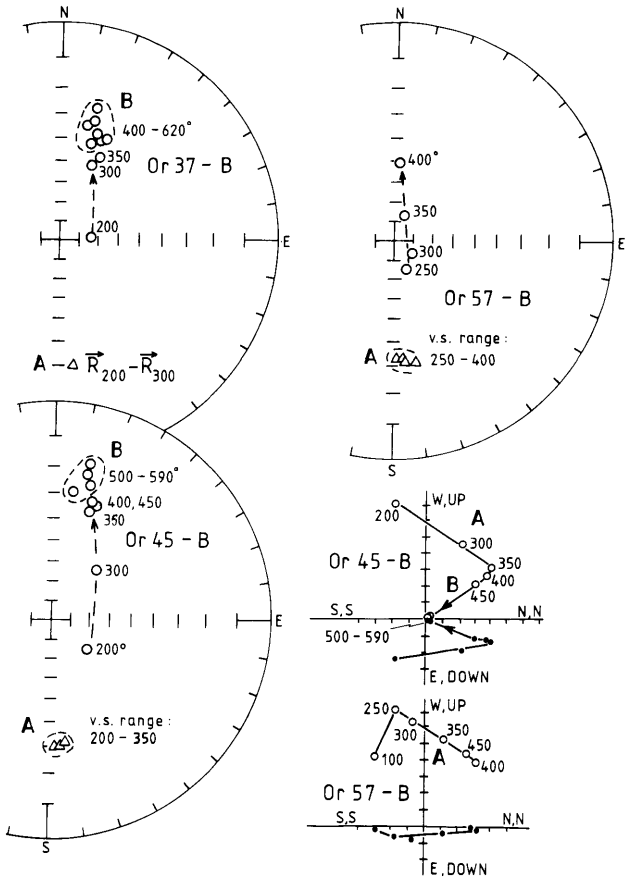


FIG. 3. Results from the baked sediments of Old Man of Hoy showing the coexisting A and B magnetizations. In the orthogonal vector diagrams the intensity unit for OR45-B is $4{\cdot}5 \times 10^{-3}$ A m^{-1} and for OR57-B it is $0{\cdot}5 \times 10^{-3}$ A m^{-1}. Diagram conventions as for Figure 2. Directions are without tectonic adjustment.

end-point directions from the flow and from the sediments are in accord with the so-called B-magnetization previously described from ORS rocks of the Caithness/Orkney region (see below).

A second magnetization, directed nearly due south and with a *c.* 30 degrees upward (negative) inclination, appears to be invariably present in the sediments. This component has blocking temperatures below *c.* 400° C and its importance increases away from the contact. However, due to component overlap with the B-magnetization, many specimens did not give a satisfactory determination of this latter magnetization which is termed the A-remanence (following the general palaeomagnetic terminology of the region).

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The remanent magnetization of lava specimens is frequently followed to well above 600° C, suggesting the presence of haematite (identified microscopically) as an important remanence carrier. A second magnetic phase of major significance has Curie temperatures (T_c) at around 560–580° C on heating, but on subsequent cooling a substantial reduction in saturation magnetization (J_s) generally takes place along with a lowering of the Curie point (sometimes by as much as 250° C). These observations and the microscopic evidence for severe low temperature alteration suggest the presence of a cation-deficient spinel (titanomaghemite). The fairly advanced state of oxidation of the iron-titanium oxides is also demonstrated by relatively high remanence coercive forces ($H_{CR} \sim 55$ mT).

$J_s - T$ curves for sediment samples show a much more linear decay pattern than for lava specimens, and the T_c for haematite (680° C) is well defined. However, the essential part of the remanent magnetization in these beds has blocking temperatures (T_B) in the 400–550° C interval suggesting the presence of magnetite/maghemite. An observed decrease in saturation magnetization away from the contact may indicate that the "magnetite" component formed during baking. A possible thermal effect is also noted in the NRM intensity (J_n) distribution: J_n is around 10–15. 10^{-3} A m^{-1} at or near the contact while the more distant sites (#8–10) have J_n values below 5 . 10^{-3} A m^{-1}.

Rackwick

While the Old Man of Hoy (OMH) lava was found to possess relatively high coercive forces rendering the AF method rather unsuitable for an appropriate demagnetization analysis, this technique has been the most successful one for the Rackwick flow. This lava is characterized by having fairly low magnetic stability both for thermal and AF demagnetization. Stable end-point results have been determined from 15 specimens of which 10 are based on the AF method. The stable directions correspond to the B-magnetization and they are all of reverse polarity (Fig. 4).

All samples showed a southerly movement into the SW-quadrant in the initial stages of demagnetization, but in a number of thermally demagnetized specimens this trend was superseded by northerly patterns into the NE-quadrant without reaching terminal directions. It is suggested therefore, that the Rackwick lava too has a two-polarity magnetization structure, but the reversed component carries the bulk of the stable remanence. This flow has the highest NRM-intensity of the collection (averaging *c.* 5 . 10^{-1} A m^{-1}). Also, the H_{CR} values are only around 15 mT, and haematite does not give an observable contribution to the remanent magnetization.

In the Rackwick ashy sediment haematite is the dominant magnetic mineral. It apparently carries fossil magnetization of both polarities, but a satisfactory separation of individual palaeomagnetic components has not been accomplished.

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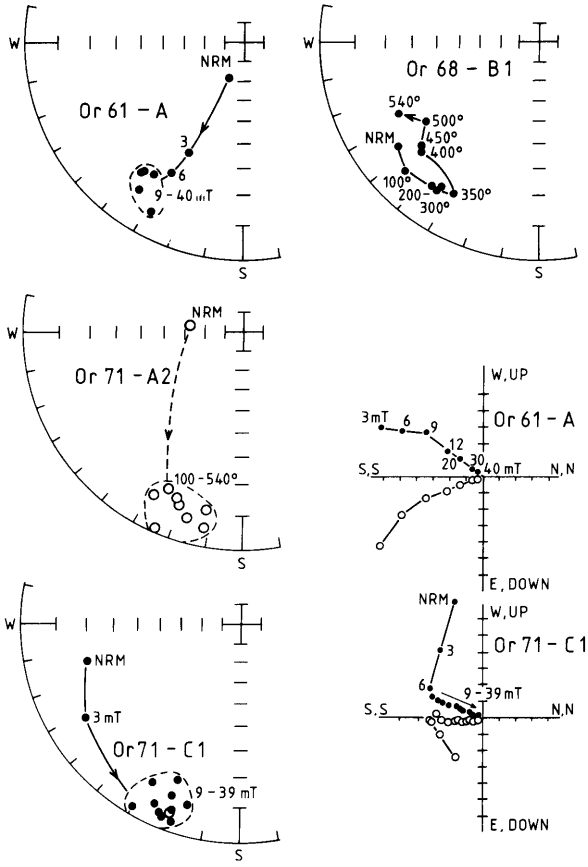


FIG. 4. Demagnetization results from the Rackwick lava flow which is apparently undisturbed tectonically. Alternating peak field values are in milli-Teslas (mT). Specimen OR68-B1 exemplifies the results of a number of specimens for which individual palaeomagnetic components could not be successfully extracted, probably consisting of superimposed magnetizations of opposite polarity. Conventions as for Figure 2.

MAGNETIZATION HISTORY AND GEOLOGICAL EVENTS

The present investigation has provided a total of 46 individual remanence directions from 41 different samples. The data are listed in Table 2. As seen from Figure 5 the directions form three coherent groupings. The B2-population (mainly from Rackwick) matches perfectly with previous results from the Orkney volcanics, the John O'Groats Sandstone, Duncansby Volcanic Neck, and the Lower Caithness beds (cf. Table 3). By applying structural correction (simple rotation of bedding about present strike direction) the B1 group (Old Man of Hoy) moves into better agreement with the regional B-axis magnetization, but it is still 20 degrees (in inclination) away from being antipodal to the B2 group. This axis

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TABLE 2

Palaeomagnetic specimen directions from the Island of Hoy along with corresponding remanence groups (A or B1,2) and associated T/AF ranges (vector subtraction ranges are marked by v.s.). A-group results are before and B-group directions (B1) are after structural correction (cf. text). See Table 1 for geological details

Site no.	Specimen	D	I	Group	Range
1	OR 1-A	032,	−26	B1	440–635° C
2	8-A	023,	−27	B1	450–580° C
	11-A	018,	−33	B1	400–635° C
	12-A	031,	−25	B1	440–635° C
3	13-A	017,	−39	B1	NRM–660° C
	14-A	031,	−30	B1	3–70 mT + 100–400° C
	16-B	199,	19	B2	400–635° C
	18-A	206,	12	B2	585°, 635° C
	19-A	025,	−35	B1	440–550° C
5	31-B	354,	−30	B1	530–620° C
	32-Ba	189,	−30	A	v.s. 200–360° C
	b	032,	−26	B1	360–605° C
	33-A	027,	−24	B1	405–605° C
	34-B	041,	−22	B1	500–670° C
	35-B	027,	−22	B1	560°, 590° C
6	36-A	027,	−34	B1	360–605° C
	37-Ba	173,	−30	A	v.s. 200–300° C
	b	030,	−28	B1	500–620° C
	38-A	016,	−21	B1	510–575° C
	38-B	014,	−26	B1	400–625° C
	39-A	020,	−21	B1	450–530° C
7	45-Aa	174,	−30	A	v.s. 250–400° C
	b	029,	−25	B1	400–600° C
	45-Ba	175,	−32	A	v.s. 200–350° C
	b	021,	−22	B1	500–590° C
	46-Aa	170,	−35	A	v.s.200–320° C
	b	017,	−27	B1	410–545° C
	46-B	018,	−25	B1	400–540° C
8	49-A	176,	−28	A	v.s. 250–450° C
10	57-A	021,	−40	B1	300–530° C
	57-B	175,	−32	A	v.s. 250–400° C
11	60-A	206,	14	B2	300–450° C
	61-A	211,	13	B2	9–40 mT
	64-A	209,	12	B2	350–450° C
12	66-A1	214,	14	B2	9–20 mT
	66-B2	218,	6	B2	9–14 mT
	66-C2	207,	4	B2	12–20 mT
13	67-C2	230,	19	B2	3–12 mT
14	68-Al	211,	8	B2	200–460° C
	68-B2	216,	17	B2	6–30 mT
15	70-B1	195,	22	B2	15–40 mT
16	71-A1	202,	−12	B2	460–520° C
	71-A2	201,	−8	B2	100–460° C
	71-B1	204,	5	B2	22,24 mT
	71-B2	200,	1	B2	3–30 mT
	71-C1	203,	7	B2	9–39 mT

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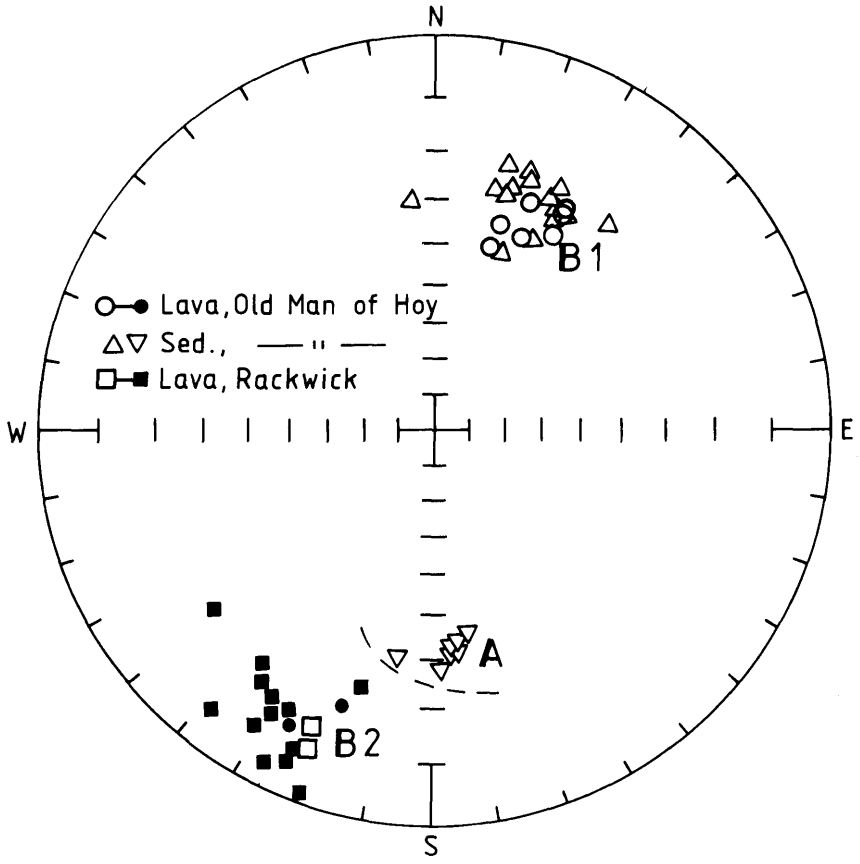


FIG. 5. Stereoplot giving the distribution of the A and B magnetizations. The characteristic (original) B-remanence forms two sub-groups, B1 and B2, that are *c.* 160° apart. The inclination difference between these two groups is attributed to tectonic activity at the Old Man of Hoy location (B1), cf. text. Directions of the B1 group are after structural correction, while no correction for the B2 population (mostly representing the tectonically undisturbed Rackwick locality) has been applied. As the essential part of the tectonic tilting in Hoy is Middle Devonian in age no correction has been applied for the superimposed A-component of Permian age.

discordance of the two groups is statistically significant at the 95 per cent level according to Watson's Test (Watson 1956).

As outlined above, the Rackwick lava shows extensive low temperature oxidation of its original Fe-Ti oxides with hardly any unaltered titanomagnetite grains left. The length of time involved in these secondary processes is certainly much longer than the "time constant" of palaeo-secular variation, a suggestion that is supported both by the evidence for a dual-polarity palaeomagnetic record, and by the excellent agreement of the mean direction of magnetization with overall

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TABLE 3

Overall palaeomagnetic results from Hoy (present study) in comparison with other relevant data. Statistics according to Fisher (1953)

Formation	Group	D	I	α_{95}	K	Pole
Hoy lavas	A	176,	−31	4·2	206	47·7N, 182·2E
(present study)	B1	023,	−28	3·8	66	
	B2	208,	9	5·5	43	22·7N, 146·3E
Orkney lavas	B	205,	8	9·1	45	23·7N, 149·5E
(Storetvedt and Petersen 1972)						
John O'Groats Sst.	A	188,	−41	3·9	32	54·2N, 163·6E
(Storetvedt and Carmichael 1979)	B	209,	6	10·4	18	24·1N, 144·5E
Duncansby Neck	A	208,	−55	2·3	155	60·3N, 126·5E
(Storetvedt *et al.* 1978)	B	205,	8	6·3	28	24·3N, 149·1E
Lower Caithness ORS	B	208,	14	13·5	26	20·5N, 147E
(Waage and Storetvedt 1973)						
Lower Caithness ORS	B	205,	3	6·4	20	27N, 148·5E
(Storetvedt and Torsvik 1983)						
Helmsdale granite; *c.* 400 m.y.	A	183,	−31	6·0	29	48·9N, 171·8E
(Torsvik *et al.* 1983)						

palaeomagnetic directions for nearby formations of similar age. The Rackwick basalt appears to have suffered only minimal tectonic disturbance. In view of the extensive low temperature oxidation encountered in these rocks, encompassing most likely very slowly acquired magnetizations, the overall palaeomagnetic directions (and associated statistics) have been based on specimen directions as unit vectors (Table 3). The average direction of group B2 (15 of its 17 samples are from the Rackwick basalt) is therefore regarded as representing the relative axial dipole field direction in Middle–Upper Devonian time.

All available rock magnetic evidence favours a more advanced stage of low temperature alteration in the OMH flow than in the Rackwick basalt. The former rock has also the most clear-cut evidence for the two-polarity magnetization build-up found in these volcanics. From the "contact test" it appears reasonable to conclude that the field had a normal polarity at the time the OHM flow cooled. From the rock magnetic evidence it seems likely that magnetite formed in the sediment during baking, but maghemite too appears to be present as a remanence carrier. Furthermore, it is recognized that the normal polarity directions of the lava are fully compatible with those of the baked sediment, while its reversely directed magnetization tends to have shallower inclination (corresponding to that of the B2 group). In view of the advanced low temperature oxidation of the Fe-Ti mineralogy found throughout the OMH lava it would be very difficult to go along with a model implying that the normally directed magnetization is essentially of deuteric origin (i.e. representing a spot reading of the field at the time of cooling), while the reversely directed remanence represents an averaging out of palaeo-secular variation owing to slow chemical acquisition processes at ambient temperatures. The

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reverse magnetization (corresponding to the B2 group) of this flow most likely post-dates the dominant normal component (B1), but it is suggested here that the discrepancy of the B1 direction (versus the regional Devonian reference field) has primarily a tectonic explanation. The tectonic model implies that prior to lava extrusion the Lower Eday beds had a local structural dip of about 20 degrees in an easterly direction before the strata, after lava cooling and subsequent low temperature oxidation during the original normal field polarity, underwent further tectonic disturbances that rotated the bedding plane of the Lower Eday sediments into its present position: a shallow westerly dip. After termination of the tectonic event the field polarity turned to reversed, and the chemical magnetization subsequently acquired, i.e. the B2 and A components, has essentially remained unaffected by tectonism. From the available geological literature on Hoy (for example Wilson *et al.* 1935) one gets the impression that the structural deformation of the Middle Devonian strata had already taken place by the time of volcanism, but the palaeomagnetic results from the Old Man of Hoy basalt suggest that volcanism and stratal disturbances were interrelated phenomena. This conclusion is geologically very reasonable.

From the discussion above, it may follow that the Rackwick flow postdates the Old Man of Hoy basalt, having formed after the tectonic phase had ceased. It is tentatively suggested therefore, that the two basalts are not true time equivalent units; i.e. they do not represent separate branches of a more extensive flow. However, this conclusion can only be given the necessary substance if a future contact test in the Rackwick ashy deposit shows clear evidence of reverse field polarity at the time of baking.

The third palaeomagnetic group, the A-component, has an axis of magnetization (and polarity) that corresponds to a Permian age. Similar secondary remanences have been established in various rocks of the Caithness region, and notably the Helmsdale granite has an A-magnetization which is in nearly perfect agreement with that of the present study (Table 3). The late Palaeozoic magnetic overprinting is not seen in the Hoy volcanics. This is attributed partly to: (a) the relatively compact nature of these rocks compared to that of the sediments, and partly to the fact that (b) they had already been through extensive low temperature oxidation in ORS time, rendering them magnetically resistant to subsequent mild hydrothermal activity. On the other hand, the established overprinting event appears to have important implications for radiometric age determination of the basalts.

Halliday *et al.* (1977) have carried out conventional K/Ar age determination on four samples of lava from Hoy, arriving at mean sample ages varying between 300 and 333 Ma. These dates are clearly too low for Devonian rocks, and the authors suggest that argon loss must have taken place during subsequent Permo-Carboniferous hydrothermal events. On the other hand, stepheating ^{40}Ar–^{39}Ar analysis for two samples (one of them from the Rackwick flow) define age plateaus of 358 and 376 Ma respectively. The authors suggest a crystallization age of around

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370 Ma (i.e. Middle Devonian). The conclusions of Halliday *et al.* (1977) are fully compatible with the present palaeomagnetic results. The B-component which constitutes the principal magnetization of the lavas matches with corresponding results for the Middle Devonian sediments of Caithness, and the evidence for argon loss through subsequent hydrothermal activity (giving too low K/Ar ages) gains support from the presence of a Permian magnetization overprint.

Discussion and conclusion

The major reason why Halliday *et al.* (1977) proposed a Permo-Carboniferous age for the overprinting event is probably that the camptonite and monchiquite dykes of the Orkney, in part cutting the Hoy Sandstone, have given Lower Permian K/Ar ages (Flett Brown 1975; Halliday *et al.* 1977). However, the question arises whether the dykes have been radiometrically reset too. Preliminary palaeomagnetic results (Storetvedt and Otterå, unpublished data) suggest indeed that this is the case, and that the true age of these intrusions may be as old as late Middle Devonian.

Another age controversy has been on the Duncansby Volcanic Neck of NE Caithness (Fig. 1). This rock has an optically very fresh mineralogy making it susceptible to acquiring secondary magnetizations during thermal-thermochemical overprinting events. From whole-rock K/Ar studies Macintyre *et al.* (1981) have arrived at a lowermost Permian age for the intrusion, while Storetvedt *et al.* (1978), combining palaeomagnetism and K/Ar studies, have suggested a Devonian–Lower Carboniferous age. The palaeomagnetic resetting of the Neck is strong, but a weak underlying remanence component has been determined. This latter magnetization accords well with the Devonian palaeomagnetic direction as established in ORS sediments of the region (the B-remanences of Table 3), and it is totally incompatible with a Permian age of the Neck.

Upsetting of radiometric dates due to relatively mild overprinting events is most likely a general problem in radiometric age determination. Complete age re-settings are probably rare so that neither the age of crystallization nor that of the overprinting (but rather a mixed age) will be obtained. Palaeomagnetism may provide very useful additional information in such cases, and the two types of study should therefore be combined wherever possible.

The Duncansby Neck along with the John O'Groats Sandstone have secondary magnetizations (A-components) that are more steeply inclined than those contaminating the Hoy sediments and the Helmsdale granite, corresponding palaeomagnetically to a Mesozoic rather than to a Permian age. This conclusion is further substantiated by the presence of only reverse polarity for the Hoy and Helmsdale overprints (corresponding to the Kiaman period from *c.* 300 Ma to the Upper Permian), while the John O'Groats/Duncansby overprints have mixed polarities

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(characterizing subsequent periods). Movement on the Great Glen Fault in mid-Mesozoic time has been established from marine geophysical studies in the Moray Firth Basin (Bacon and Chesher 1975), and we suggest that, due to the high porosity of the John O'Groats Sandstone and the magnetically susceptible nature of the Neck, these rocks also became remagnetized in the Jurassic during which any previously acquired secondary magnetization (of Permian age) was strongly replaced/overprinted.

A tectonic phase of Middle Devonian age is clearly established in Hoy but in the John O'Groats Sandstone the tectonism may locally postdate the inferred magnetization overprint (Storetvedt and Carmichael 1979). It appears reasonable to conclude therefore, that the integral structural deformation of the Devonian strata of Orkney/Caithness includes both mid-Devonian and Permo-Mesozoic tectonic components. These phases of deformation were probably related to tectonic reactivation of the Great Glen Fault.

From the present study of rocks from Hoy the following conclusions can be drawn:

(1) The Middle Devonian Hoy lavas possess a well-defined principal component of magnetization with NNE-SSW declination and shallow inclination. The inferred relative palaeomagnetic pole is at 146·3E, 22·7N (present grids) which is in extremely good agreement with results from the Middle Devonian sediments of Caithness. The consistent shallow palaeomagnetic inclination demonstrates very clearly that in the Middle Devonian, Scotland west of the Great Glen Fault had European palaeolatitudes (within the palaeomagnetic accuracy of a few degrees), dismissing therefore the extravagant palaeolatitude displacement along the Fault as suggested by Van der Voo and Scotese (1981).

(2) Detailed consideration of palaeomagnetic directions suggests that the Hoy volcanism was linked to the tectonic deformation event of the underlying Lower Eday beds.

(3) It is tentatively concluded that the Rackwick and Old Man of Hoy basalts do not represent branches of the same lava flow.

(4) In Permian time the area of Hoy suffered a relatively mild hydrothermal event impressing a second palaeomagnetic component into the sediments. The lavas did not acquire this additional magnetization (owing to their fairly compact nature and stable magneto-mineralogical state); but on the other hand, radiometric studies (Halliday *et al.* 1977) suggest that the hydrothermal overprinting event led to loss of argon resulting in too low K/Ar ages of these rocks. Thus, the geological inferences drawn from the radiometric and palaeomagnetic studies are in full correspondence.

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